



STUDY BUILD
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PROJECT 02

CUSTOMER ANALYTICS

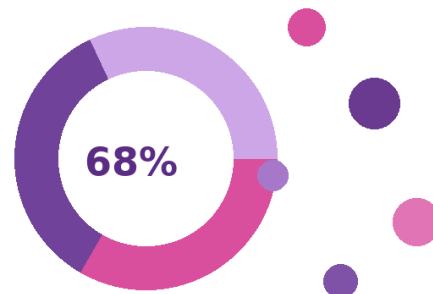
From Data to Business Decisions

Customer Performance Dashboard

Customers
1,248

Revenue
\$84.2K

Satisfaction
4.6 / 5



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About This Project

In this project, you will work as a Junior Data Analyst for an online retail company. The CEO is not looking for a simple statistical report. The goal is to understand what decisions the data supports and to convert analysis into practical business recommendations.

Each section begins with a realistic management request. It then identifies the required variables, suggests suitable visualizations, provides a complete Python example, explains the main parts of the code, and describes the expected analytical output.

Project objective

Understand the business question, select an appropriate analytical approach, communicate evidence clearly, and recommend an action that can be justified with the available data.

Important Note

The code examples in this booklet are provided only as learning aids. They were prepared with AI assistance to illustrate one possible workflow.

- Review and test the code before using it. Correct or adapt any part that does not match your data or working environment.
- Using the sample code is optional. You may complete the project with Python, R, Excel, Power BI, SQL, or any other suitable tool.
- The main goal is to understand the problem, choose a reasonable method, and interpret the results correctly - not to copy a specific code solution.
- Your submission does not need to be perfect or fully complete. A thoughtful partial analysis is better than copied output that you cannot explain.

Assessment priority

The quality of your reasoning, interpretation, and recommendations is more important than the programming language or software you use.

Dataset Description

Each row represents a summarized customer profile. The dataset does not contain a separate row for every order or every purchased item.

Column	Meaning
customer_id	Unique customer identifier
first_name	Customer first name
gender	Customer gender
age	Customer age
city	Customer city
province	Customer province
signup_date	Customer registration date
membership_tier	Customer membership level
purchase_count	Number of purchases made by the customer
avg_order_value	Average order value
total_spending	Total amount spent by the customer
last_purchase_days	Number of days since the customer's most recent purchase
payment_method	Customer payment method
device	Device used by the customer
discount_used	Whether the customer used a discount at least once
returned_items	Number of returned items
satisfaction_score	Customer satisfaction score

Important interpretation notes

- The reference date used to calculate `last_purchase_days` is not provided. You may compare customer recency, but you cannot calculate the exact calendar date of the last purchase unless you explicitly define an assumed reference date.
- `returned_items` represents the number of returned items, not the number of returned orders.
- `discount_used` indicates whether a customer used a discount at least once in the recorded data. It does not describe only the most recent purchase.

Initial Data Preparation and File Checks

Place the Excel file in the same folder as your Notebook. The following code checks whether the file exists, verifies the required columns, converts the registration date to a proper `DateTime` format, converts numerical fields, trims extra spaces from text fields, and reports basic data-quality issues.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path

FILE_PATH = Path("customer_data.xlsx")

if not FILE_PATH.exists():
    raise FileNotFoundError(f"File not found: {FILE_PATH}")

df = pd.read_excel(FILE_PATH)

required_columns = [
    "customer_id", "first_name", "gender", "age", "city", "province",
```

```

"signup_date", "membership_tier", "purchase_count", "avg_order_value",
"total_spending", "last_purchase_days", "payment_method", "device",
"discount_used", "returned_items", "satisfaction_score"
]

missing_columns = [col for col in required_columns if col not in df.columns]
if missing_columns:
    raise ValueError(f"Required columns are missing: {missing_columns}")

df["signup_date"] = pd.to_datetime(df["signup_date"], errors="coerce")

numeric_columns = [
    "age", "purchase_count", "avg_order_value", "total_spending",
    "last_purchase_days", "returned_items", "satisfaction_score"
]

for col in numeric_columns:
    df[col] = pd.to_numeric(df[col], errors="coerce")

text_columns = [
    "first_name", "gender", "city", "province", "membership_tier",
    "payment_method", "device", "discount_used"
]

for col in text_columns:
    df[col] = df[col].astype("string").str.strip()

print("Dataset shape:", df.shape)
print("Duplicate rows:", df.duplicated().sum())
print("Missing values by column:")
print(df.isna().sum().sort_values(ascending=False))

display(df.head())

```

Code element	Purpose
import pandas as pd	Reads Excel files, manages data frames, groups records, and calculates analytical summaries.
import numpy as np	Supports numerical operations and special numerical values.
import matplotlib.pyplot as plt	Creates the charts used throughout the project.
Path("customer_data.xlsx")	Defines the dataset location in a platform-independent way.
FILE_PATH.exists()	Confirms that the file exists before attempting to read it.
pd.read_excel(...)	Loads the Excel file into a pandas DataFrame.
required_columns	Lists the exact variables needed by the project.
missing_columns	Identifies required variables that are not present in the file.
pd.to_datetime(..., errors="coerce")	Converts valid dates and replaces invalid date strings with missing values.
pd.to_numeric(..., errors="coerce")	Converts numeric fields and changes invalid values to missing values.
astype("string").str.strip()	Converts text fields to string format and removes extra spaces.
df.isna().sum()	Counts missing values in every column.

Before continuing

- If missing values are found, explain how you handled them in README.md.
- If duplicate records are found, document how you identified them and why you removed or kept them.
- Record every data-type correction, unreasonable-value correction, and cleaning decision in README.md.

QUESTION 1

Who Are the Store's Core Customers?

CEO request

Before designing a marketing campaign, I need a clear profile of our current customers. Examine age groups, gender, membership level, device, and payment method. Identify the dominant customer profile and explain what it means for our communication strategy.

Required variables

- customer_id
- gender
- age
- membership_tier
- device
- payment_method

Expected visualizations

- Bar chart showing the number of customers in each age group
- Bar chart showing membership-tier distribution
- Bar chart showing device distribution
- Bar chart showing payment-method distribution

Python example

```
analysis_df = df.dropna(
    subset=["customer_id", "gender", "age", "membership_tier", "device", "payment_method"]
).copy()

age_bins = [0, 24, 34, 44, 54, 64, np.inf]
age_labels = ["Under 25", "25-34", "35-44", "45-54", "55-64", "65+"]

analysis_df["age_group"] = pd.cut(
    analysis_df["age"],
    bins=age_bins,
    labels=age_labels,
    include_lowest=True
)

profile_summary = {
    "Number of customers": analysis_df["customer_id"].nunique(),
    "Mean age": analysis_df["age"].mean(),
    "Median age": analysis_df["age"].median(),
    "Most common gender": analysis_df["gender"].mode().iloc[0],
    "Most common membership tier": analysis_df["membership_tier"].mode().iloc[0],
    "Most common device": analysis_df["device"].mode().iloc[0],
    "Most common payment method": analysis_df["payment_method"].mode().iloc[0]
}

display(pd.Series(profile_summary, name="Value"))

age_counts = analysis_df["age_group"].value_counts(sort=False)
age_counts.plot(kind="bar", figsize=(9, 5))
plt.title("Customers by Age Group")
plt.xlabel("Age group")
plt.ylabel("Number of customers")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

for column, title in [
    ("membership_tier", "Customer Membership-Tier Distribution"),
    ("device", "Devices Used by Customers"),
    ("payment_method", "Customer Payment Methods")
]:
```

```
counts = analysis_df[column].value_counts().sort_values(ascending=True)
counts.plot(kind="barh", figsize=(8, 4.5))
plt.title(title)
plt.xlabel("Number of customers")
plt.ylabel("")
plt.tight_layout()
plt.show()
```

Code walkthrough

Code element	Purpose
dropna(subset=...)	Keeps only records containing the variables required for this question.
pd.cut(...)	Transforms continuous age values into business-friendly age groups.
nunique()	Counts unique customers.
mean() / median()	Summarizes the center of the age distribution from two perspectives.
mode().iloc[0]	Returns the most frequent category.
value_counts()	Counts the number of customers in each category.
plot(kind="bar")	Compares age-group frequencies.
plot(kind="barh")	Improves readability when category labels are long.
tight_layout()	Prevents titles and axis labels from being cut off.

Expected business output

- A summary table of the dominant customer characteristics
- Four clearly labeled charts
- One paragraph defining the dominant customer persona using only the available evidence

Business interpretation

Which age group is largest? Which membership level, device, and payment method are dominant? Should the next campaign focus only on the largest segment, or could a smaller but more valuable segment emerge in the later questions?

QUESTION 2

Which Province and City Should Receive More Advertising Investment?

CEO request

Our advertising budget is limited. Identify the province and city that should receive priority. Do not decide using total sales alone; also consider customer count, average spending, purchase behavior, and satisfaction.

Required variables

- province
- city
- customer_id
- total_spending
- purchase_count
- avg_order_value
- satisfaction_score

Expected visualizations

- Horizontal bar chart of the top 10 cities by total customer spending
- Scatter plot of city revenue versus average satisfaction, with point size representing customer count

Python example

```
location_df = df.dropna(
    subset=["province", "city", "customer_id", "total_spending", "satisfaction_score"]
).copy()

city_summary = (
    location_df.groupby(["province", "city"], as_index=False)
    .agg(
        customer_count=("customer_id", "nunique"),
        total_revenue=("total_spending", "sum"),
        avg_customer_spending=("total_spending", "mean"),
        avg_purchase_count=("purchase_count", "mean"),
        avg_order_value=("avg_order_value", "mean"),
        avg_satisfaction=("satisfaction_score", "mean")
    )
)

city_summary = city_summary[city_summary["customer_count"] >= 3].copy()

top_cities = city_summary.nlargest(10, "total_revenue").sort_values("total_revenue")
display(top_cities.sort_values("total_revenue", ascending=False))

labels = top_cities["province"] + " - " + top_cities["city"]
plt.figure(figsize=(10, 6))
plt.barh(labels, top_cities["total_revenue"])
plt.title("Top 10 Cities by Total Customer Spending")
plt.xlabel("Total customer spending")
plt.ylabel("Province - city")
plt.tight_layout()
plt.show()

plt.figure(figsize=(9, 6))
plt.scatter(
    city_summary["total_revenue"],
    city_summary["avg_satisfaction"],
    s=city_summary["customer_count"] * 20,
    alpha=0.65
)
plt.title("City Financial Value vs. Customer Satisfaction")
plt.xlabel("Total customer spending")
plt.ylabel("Average satisfaction")
```

```
plt.grid(alpha=0.25)
plt.tight_layout()
plt.show()

recommended_city = city_summary.sort_values(
    ["total_revenue", "avg_satisfaction", "customer_count"],
    ascending=[False, False, False]
).head(1)

display(recommended_city)
```

Code walkthrough

Code element	Purpose
groupby(["province", "city"])	Groups customers by both province and city so that cities with the same name are not combined incorrectly.
.agg(...)	Calculates several management indicators for each city.
customer_count=(..., "nunique")	Counts unique customers in each city.
total_revenue=(..., "sum")	Calculates total spending generated by the city's customers.
customer_count >= 3	Removes extremely small groups that may produce unstable averages. Document this threshold.
nlargest(10, "total_revenue")	Selects the ten cities with the highest total spending.
scatter(...)	Displays the relationship between financial value and satisfaction.
s=customer_count * 20	Scales each point according to the number of customers.
sort_values([...])	Produces an initial recommendation using revenue, satisfaction, and customer count.

Expected business output

- A city-level table containing customer count, total revenue, average spending, purchase behavior, and satisfaction
- One recommended city and one alternative city
- A clear explanation of the advantage and risk associated with each choice

Business interpretation

Is the leading city ranked highly only because it has more customers, or is average customer spending also high? Is satisfaction strong enough to support further investment? Justify the final recommendation with at least three indicators.

QUESTION 3

Which Customers Should Be Prioritized for the Loyalty Program?

CEO request

We can include only ten customers in the loyalty program. Select customers who combine strong financial value, frequent purchasing, recent activity, and good satisfaction. Membership tier alone is not enough.

Required variables

- customer_id
- first_name
- membership_tier
- purchase_count
- total_spending
- last_purchase_days
- satisfaction_score

Expected visualization

- Scatter plot of purchase count versus total spending, with more recently active customers shown as larger points

Python example

```
loyalty_df = df.dropna(
    subset=[
        "customer_id", "first_name", "purchase_count", "total_spending",
        "last_purchase_days", "satisfaction_score"
    ]
).copy()

for column in ["purchase_count", "total_spending", "satisfaction_score"]:
    minimum = loyalty_df[column].min()
    maximum = loyalty_df[column].max()

    if maximum == minimum:
        loyalty_df[f"{column}_score"] = 1.0
    else:
        loyalty_df[f"{column}_score"] = (
            loyalty_df[column] - minimum
        ) / (maximum - minimum)

recency_min = loyalty_df["last_purchase_days"].min()
recency_max = loyalty_df["last_purchase_days"].max()

if recency_max == recency_min:
    loyalty_df["recency_score"] = 1.0
else:
    loyalty_df["recency_score"] = 1 - (
        (loyalty_df["last_purchase_days"] - recency_min)
        / (recency_max - recency_min)
    )

loyalty_df["loyalty_score"] = (
    0.35 * loyalty_df["total_spending_score"]
    + 0.30 * loyalty_df["purchase_count_score"]
    + 0.20 * loyalty_df["recency_score"]
    + 0.15 * loyalty_df["satisfaction_score_score"]
)

top_loyal_customers = loyalty_df.nlargest(10, "loyalty_score")[[
    "customer_id", "first_name", "membership_tier", "purchase_count",
    "total_spending", "last_purchase_days", "satisfaction_score",
    "loyalty_score"
]]
```

```

display(top_loyal_customers)

plt.figure(figsize=(10, 6))
plt.scatter(
    loyalty_df["purchase_count"],
    loyalty_df["total_spending"],
    s=(loyalty_df["recency_score"] + 0.15) * 140,
    alpha=0.6
)
plt.scatter(
    top_loyal_customers["purchase_count"],
    top_loyal_customers["total_spending"],
    s=180,
    marker="*",
    label="Top 10 selected customers"
)
plt.title("Purchase Count vs. Total Customer Spending")
plt.xlabel("Purchase count")
plt.ylabel("Total spending")
plt.legend()
plt.grid(alpha=0.25)
plt.tight_layout()
plt.show()

```

Code walkthrough

Code element	Purpose
Min-max normalization	Transforms variables measured on different scales to values between zero and one.
if maximum == minimum	Prevents division by zero when every customer has the same value.
1 - normalized last_purchase_days	Reverses the recency measure so that a more recent customer receives a higher score.
0.35, 0.30, 0.20, 0.15	Weights total spending, purchase frequency, recency, and satisfaction. State these weights as analytical assumptions.
nlargest(10, "loyalty_score")	Selects the ten highest-scoring customers.
marker="*"	Highlights selected customers on the scatter plot.

Expected business output

- A ranked table of ten selected customers and their loyalty scores
- An explanation of the weighting logic
- At least one evidence-based reason for selecting each customer

Business interpretation

Do selected customers simultaneously have high purchase counts and high total spending? Should a customer with high historical spending but very old activity still receive a reward? Compare the calculated score with the customer's current membership tier and explain inconsistencies.

QUESTION 4

Which Valuable Customers Are at Risk of Becoming Inactive?

CEO request

Identify customers who previously created strong value but have not purchased recently. We want to design a reactivation campaign before these customers are completely lost.

Required variables

- customer_id
- first_name
- total_spending
- purchase_count
- last_purchase_days
- membership_tier
- satisfaction_score

Expected visualization

- Scatter plot of days since last purchase versus total spending, with at-risk customers clearly highlighted

Python example

```
risk_df = df.dropna(
    subset=[
        "customer_id", "first_name", "total_spending",
        "purchase_count", "last_purchase_days"
    ]
).copy()

spending_threshold = risk_df["total_spending"].quantile(0.75)
purchase_threshold = risk_df["purchase_count"].median()
inactivity_threshold = risk_df["last_purchase_days"].quantile(0.75)

risk_df["at_risk"] = (
    (risk_df["total_spending"] >= spending_threshold)
    & (risk_df["purchase_count"] >= purchase_threshold)
    & (risk_df["last_purchase_days"] >= inactivity_threshold)
)

at_risk_customers = risk_df.loc[risk_df["at_risk"], [
    "customer_id", "first_name", "membership_tier", "purchase_count",
    "total_spending", "last_purchase_days", "satisfaction_score"
]].sort_values(
    ["total_spending", "last_purchase_days"],
    ascending=[False, False]
)

display(at_risk_customers)
print("Number of at-risk customers:", len(at_risk_customers))

plt.figure(figsize=(10, 6))
plt.scatter(
    risk_df.loc[~risk_df["at_risk"], "last_purchase_days"],
    risk_df.loc[~risk_df["at_risk"], "total_spending"],
    alpha=0.45,
    label="Other customers"
)
plt.scatter(
    risk_df.loc[risk_df["at_risk"], "last_purchase_days"],
    risk_df.loc[risk_df["at_risk"], "total_spending"],
    marker="X",
    s=110,
    label="Valuable customers at risk"
)
```

```
plt.axvline(inactivity_threshold, linestyle="--", label="Inactivity threshold")
plt.axhline(spending_threshold, linestyle="--", label="Value threshold")
plt.title("Valuable Customers at Risk of Becoming Inactive")
plt.xlabel("Days since last purchase")
plt.ylabel("Total customer spending")
plt.legend()
plt.grid(alpha=0.25)
plt.tight_layout()
plt.show()
```

Code walkthrough

Code element	Purpose
quantile(0.75)	Identifies the top 25% of customers for financial value and the top 25% for inactivity.
median()	Creates a purchase-frequency threshold that is less sensitive to extreme values than the mean.
&	Requires all three conditions to be true before a customer is classified as at risk.
loc[risk_df["at_risk"]]	Selects only customers who meet the risk definition.
axvline / axhline	Displays the decision thresholds directly on the chart.
sort_values([...])	Prioritizes customers with higher value and greater inactivity.

Expected business output

- A clear, data-based definition of a valuable customer at risk
- A prioritized reactivation list
- Different recommended actions for dissatisfied customers and satisfied but inactive customers

Business interpretation

How many customers appear in the high-value, high-inactivity region? Are they dissatisfied, or have they simply not purchased recently? Which customers should receive immediate contact or a personalized offer? The dataset supports an inactivity-risk interpretation, not proof of confirmed churn.

QUESTION 5

How Are Discounts, Devices, and Payment Methods Associated with Customer Behavior?

CEO request

Managers disagree about discounts. Some believe discounts increase purchasing, while others are concerned about returns. Compare customers by discount use, device, and payment method. Keep the analysis descriptive and do not claim a causal effect.

Required variables

- discount_used
- device
- payment_method
- total_spending
- purchase_count
- avg_order_value
- returned_items
- satisfaction_score

Expected visualizations

- Bar chart of average total spending by discount-use group
- Bar chart of average returned items by discount-use group
- Bar chart of average total spending by device
- Bar chart of average total spending by payment method

Python example

```
factor_df = df.dropna(
    subset=[
        "customer_id", "discount_used", "device", "payment_method",
        "total_spending", "purchase_count", "avg_order_value",
        "returned_items", "satisfaction_score"
    ]
).copy()

summary_by_discount = (
    factor_df.groupby("discount_used", as_index=False)
    .agg(
        customer_count=("customer_id", "nunique"),
        avg_total_spending=("total_spending", "mean"),
        avg_purchase_count=("purchase_count", "mean"),
        avg_order_value=("avg_order_value", "mean"),
        avg_returned_items=("returned_items", "mean"),
        avg_satisfaction=("satisfaction_score", "mean")
    )
)

display(summary_by_discount)

metrics = [
    ("avg_total_spending", "Average Total Spending by Discount Use", "Average total spending"),
    ("avg_returned_items", "Average Returned Items by Discount Use", "Average returned items")
]

for metric, title, ylabel in metrics:
    summary_by_discount.plot(
        x="discount_used",
        y=metric,
        kind="bar",
        legend=False,
        figsize=(7, 4.5)
```

```

)
plt.title(title)
plt.xlabel("Discount used")
plt.ylabel(ylabel)
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

for factor, title in [
    ("device", "Average Total Spending by Device"),
    ("payment_method", "Average Total Spending by Payment Method")
]:
    factor_summary = (
        factor_df.groupby(factor)["total_spending"]
        .agg(["count", "mean"])
        .query("count >= 3")
        .sort_values("mean", ascending=True)
    )

    display(factor_summary)
    factor_summary["mean"].plot(kind="barh", figsize=(8, 4.5))
    plt.title(title)
    plt.xlabel("Average total customer spending")
    plt.ylabel("")
    plt.tight_layout()
    plt.show()

```

Code walkthrough

Code element	Purpose
groupby("discount_used")	Compares customers who used discounts with customers who did not.
.agg(...)	Calculates financial value, purchase frequency, order value, returns, satisfaction, and group size.
for metric, title, ylabel in metrics	Creates two discount charts without repeating the plotting code.
groupby(factor)["total_spending"]	Groups total customer spending by device or payment method.
agg(["count", "mean"])	Calculates both group size and average spending.
query("count >= 3")	Removes extremely small categories from the main comparison.
barh	Creates a readable ranked comparison for devices and payment methods.

Expected business output

- A comparison table for customers with and without recorded discount use
- A description of the observed pattern between discount use, spending, purchase frequency, returns, and satisfaction
- Identification of the device and payment method with the highest average customer value
- The explicit statement: "This analysis does not establish a causal relationship."

Business interpretation

Do customers who used discounts spend more? Do they also return more items? How many customers are included in each comparison? Which device or payment method has the highest average, and is the sample size large enough to support a business decision?

QUESTION 6

Build a Five-Minute Executive Report

CEO request

I have only five minutes to review the analysis. Create a one-page executive report showing the overall state of the business, the most important customer opportunity, and the most important customer risk. Use no more than six KPIs and only three charts.

Data used

This section reuses the validated outputs from the previous questions and may use any variable in the dataset.

Expected visual output

- A summary table containing no more than six KPIs
- Three selected charts from the earlier questions
- Three management recommendations using the structure KPI → Evidence → Action

Python example

```
kpi_summary = pd.Series({
    "Unique customers": df["customer_id"].nunique(),
    "Total customer spending": df["total_spending"].sum(),
    "Average spending per customer": df["total_spending"].mean(),
    "Average purchase count": df["purchase_count"].mean(),
    "Average satisfaction": df["satisfaction_score"].mean(),
    "Total returned items": df["returned_items"].sum()
}, name="Value")

display(kpi_summary)

# Chart 1: Top 10 cities
chart1 = city_summary.nlargest(10, "total_revenue").sort_values("total_revenue")
chart1_labels = chart1["province"] + " - " + chart1["city"]

plt.figure(figsize=(10, 6))
plt.barh(chart1_labels, chart1["total_revenue"])
plt.title("Top 10 Cities by Total Customer Spending")
plt.xlabel("Total customer spending")
plt.tight_layout()
plt.show()

# Chart 2: Valuable customers at risk
plt.figure(figsize=(10, 6))
plt.scatter(
    risk_df.loc[~risk_df["at_risk"], "last_purchase_days"],
    risk_df.loc[~risk_df["at_risk"], "total_spending"],
    alpha=0.4,
    label="Other customers"
)
plt.scatter(
    risk_df.loc[risk_df["at_risk"], "last_purchase_days"],
    risk_df.loc[risk_df["at_risk"], "total_spending"],
    marker="x",
    s=110,
    label="At risk"
)
plt.title("Valuable Customers at Risk of Becoming Inactive")
plt.xlabel("Days since last purchase")
plt.ylabel("Total spending")
plt.legend()
plt.tight_layout()
plt.show()

# Chart 3: Discount comparison
summary_by_discount.plot(
    x="discount_used",
    y=["avg_total_spending", "avg_returned_items"],
    kind="bar",
```

```

figsize=(9, 5)
)
plt.title("Customer Value and Returns by Discount Use")
plt.xlabel("Discount used")
plt.ylabel("Average")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

# Export core result tables
with pd.ExcelWriter("customer_analytics_results.xlsx") as writer:
    kpi_summary.to_excel(writer, sheet_name="KPI")
    city_summary.to_excel(writer, sheet_name="City Analysis", index=False)
    top_loyal_customers.to_excel(writer, sheet_name="Top Loyal Customers", index=False)
    at_risk_customers.to_excel(writer, sheet_name="At Risk Customers", index=False)
    summary_by_discount.to_excel(writer, sheet_name="Discount Analysis", index=False)

print("customer_analytics_results.xlsx has been saved.")

```

Code walkthrough

Code element	Purpose
pd.Series({...})	Places the principal KPIs in a simple two-column management table.
nunique()	Calculates the true number of unique customers.
sum() / mean()	Summarizes total activity and average customer performance.
Reusing city_summary and risk_df	Avoids repeating calculations and keeps the final report consistent with earlier analysis.
pd.ExcelWriter(...)	Stores several output tables in separate worksheets within one Excel file.
to_excel(..., index=False)	Exports tables without the pandas row-number column.

Recommended structure for each management recommendation

- KPI: State the relevant performance measure.
- Evidence: Cite the table, chart, or comparison supporting the conclusion.
- Action: Recommend a specific and measurable business response.

Project Deliverables

- Notebook named customer_analytics_username.ipynb
- Excel output named customer_analytics_results.xlsx
- The cleaned dataset used in the analysis
- README.md describing the problem, workflow, findings, limitations, and recommendations
- Optional short PDF report containing the three final charts and three management recommendations

Suggested Submission Structure

```

submissions/
├── your-github-username/
│   ├── customer_analytics_username.ipynb
│   ├── customer_analytics_results.xlsx
│   ├── cleaned_customer_data.xlsx
│   ├── README.md
│   ├── executive_report.pdf    # optional
│   └── charts/
│       ├── chart_01.png
│       ├── chart_02.png
│       └── chart_03.png

```


Suggested README Structure

- Project Overview
- Dataset Description
- Data Quality Checks
- Business Questions
- Method and Tools
- Key Findings
- Business Recommendations
- Analysis Limitations
- Files Included

Analysis Limitations

- Each row is a summary of one customer. Individual order-level records are not available, so order-level patterns cannot be examined.
- `last_purchase_days` provides only the elapsed time since the most recent purchase. The reference date and complete purchase history are not available.
- The discount analysis is observational. Differences between groups do not prove that discounts caused higher spending, more purchases, or more returns.
- `returned_items` provides the number of returned items. Because the total number of purchased items is unavailable, the true return rate cannot be calculated.
- Averages for cities, categories, or groups with few customers may be unstable and should be interpreted cautiously.
- The loyalty score, risk thresholds, and minimum group sizes are analytical choices. They should be reported transparently and can be tested with alternative assumptions.
- Results apply only to the available dataset and should not be generalized beyond it without additional evidence.

Final Analyst Checklist

- ☐ Do all charts have clear titles and labeled axes?
- ☐ Does every conclusion include a supporting table, value, or chart?
- ☐ Have missing values, duplicates, and data corrections been documented?
- ☐ Have you distinguished association from causation?
- ☐ Have optional thresholds and weights been explained?
- ☐ Are the recommendations practical, measurable, and linked to evidence?
- ☐ Have small sample sizes been considered?
- ☐ Have the dataset limitations been reported honestly?
- ☐ Can you explain every code block you used?
- ☐ Does the final submission follow the required folder and file-naming structure?

Portfolio note

After completing this project, you can accurately state that you performed a customer analytics project for an online retail scenario. In a portfolio or interview, describe the exact tasks you completed, the tools you used, the decisions you made, and the business recommendations supported by your analysis.